

cSharp

-Paul Sherwood-

April 21, 2025

Contents

Chapter 1

1

1.1 Video 1: An Introduction to the C+ Learning Cycle

The photo on the left shows large, stalk-like saguaro cacti with multiple arms Figure 18.1 All organisms are products of evolution adapted to their environment. Chapter Outline 18.1 Understanding Evolution 18.2 Formation of New Species 18.3 Reconnection and Speciation Rates All living organisms, from bacteria to baboons to blueberries

The theory of evolution is the unifying theory of biology

Footnotes 1Theodosius Dobzhansky. "Biology, Molecular and Organismic." American Zoologist 4, no. 4 (1964): 449.

1.1.1 Understanding Evolution

Learning Objectives By the end of this section, you will be able to do the following:

Describe how scientists developed the present-day theory of evolution Define adaptation Explain convergent and divergent evolution Describe homologous and vestigial structures Discuss misconceptions about the theory of evolution Evolution by natural selection describes a mechanism for

Also during the eighteenth century, James Hutton, a Scottish

Charles Darwin and Natural Selection In the mid-nineteenth century, two naturalists, Charles Darwin

Figure 18.2 Darwin observed that beak shape varies among

Wallace and Darwin both observed similar patterns in other

For example, Darwin observed a population of giant tortoises

Natural selection, Darwin argued, was an inevitable outcome

In 1858, Darwin and Wallace (Figure 18.3) presented papers

It is difficult and time-consuming to document and present